MANSA AGGARWAL

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GitHub | LinkedIn | LeetCode | Portfolio

SUMMARY

Computer Science graduate seeking entry level roles in embedded systems, computer vision, IoT, machine learning, and AI. Builds projects end to end across real time vision, embedded hardware, and LLM tools. Interested in TinyML, and practices DSA in C++ on LeetCode.

EDUCATION

B.Tech, Computer Science *Graduated May 2026*
Maharishi Dayanand University (MDU)

COMPUTER VISION PROJECT

Gesture App | GitHub
Tech: Python, OpenCV, MediaPipe (Hands, FaceMesh), Pygame, PyAutoGUI
Problem: real time gesture control from one webcam at a usable frame rate

- Draw mode: air drawing with adjustable brush size, an eraser, and a timed 7 swatch color palette
- Count mode: finger count gestures across two hands mapped to slide controls (next, previous, start, stop) with hold based confirmation and audio feedback
- Filter mode: draw a custom face filter on a FaceMesh skeleton outline, then track it live on your face
- Tuned MediaPipe settings to balance accuracy against frame rate and documented hand orientation limits in the README

EMBEDDED AND IOT PROJECTS

ESP32 Weather Dashboard | GitHub
Tech: ESP32, DHT11 sensor, Arduino C++, on device WiFi web server
Problem: hardware bring up (wiring, firmware, drivers) and serving live readings with no external server

Traffic Light Controller with LED Matrix Countdown | GitHub
Tech: Arduino C++, 8x8 LED matrix (MAX7219)
Problem: countdown synced to each light state, using a custom digit bitmap font

SOFTWARE AND LLM PROJECTS

ATS Resume Scorer | Live Demo | GitHub
Tech: Python, Groq LLM API, Streamlit
Problem: scoring a resume against a job description, with UI and scoring logic in separate modules

Shell (CodeCrafters challenge) | GitHub
Tech: Python, os.pipe, subprocess
Problem: shell with REPL, builtins, PATH lookup, I/O redirection; pipelines in progress

DATA STRUCTURES AND ALGORITHMS

Regular practice in C++ on LeetCode, including solutions shared with the community.

SKILLS

Languages and tools: Python, C++, Git, GitHub
ML and Vision: NumPy, scikit-learn, TensorFlow, Keras, OpenCV, MediaPipe, Groq API
Embedded and Hardware: Arduino, ESP32, sensor integration, OpenWrt (router firmware)
Coursework: Data Structures and Algorithms, Discrete Mathematics, Machine Learning